



MANGALORE UNIVERSITY

**State Education Policy – 2024
[SEP-2024]**

CURRICULUM

FOR

BCA

BACHELOR OF COMPUTER APPLICATIONS

MANGALORE UNIVERSITY

Suggested Programme Structure for the Under Graduate Programmes

[BCA, BCA (A.I & M.L), BCA (D.A)]

S e m	Course 1	Course 2	Course 3	El e c t i v e / O p t i o n a l	C o u r s e	L a n g u a g e	C o m p u l s o r y	T o t a l C r e d i t	Total Working hour
I	5 (3T+2P)	5 (3T+2P)	5 T			3+3	2	23	4+4+4+4+5+4+ 4+2=31
II	5 (3T+2P)	5 (3T+2P)	5T			3+3	2	23	4+4+4+4+5+4+ 4+2=31
III	5 (3T+2P)	5 (3T+2P)	5T	2		3+3		23	4+4+4+4+5+4+ 4+2=31
IV	5 (3T+2P)	5 (3T+2P)	5T	2		3+3	2	25	4+4+4+4+5+2+ 4+4+2=33
V	8[(2x3T)+2P]	8[(2x3T)+2P]	8[(2x3T)+2P]				2	26	3+3+3+3+3+3+ 4+4+4+2=32
VI	3T+2P	3T+2P	Research Methodology + Project Work- 12					22	3+3+4+4+22=36
Total Credits									142

Note:

- **Course1 and Course2: I to IV Semester: Theory 3 credit=4 contact hours & Practical 2 credit=4 contact hours**
- **Course3: I to IV Semester: Theory 5 credit=5 contact hours**

- **Course1, Course2 and Course3: V and VI Semester: Theory 3 credit=3 contact hours & Practical 2 credit=4 contact hours**
- **Elective/Optional: 2 credit=2 contact hours**
- **Languages: 3 credit=4 contact hours**
- **Compulsory: 2 credits=2 contact hours**

CURRICULUM STRUCTURE FOR V TO VI SEMETER BCA

Semester V								
Sl. No	Course Code	Title of the Course	Category of Courses	Teaching Hours per Week	SEE	IA	Total Marks	Credits
1	BCACAC S501	Software Engineering	Core	3	80	20	100	3
2	BCACAC S502	Dynamic Web Programming	Core	3	80	20	100	3
3	BCACAC S503	Design and Analysis of Algorithms	Core	3	80	20	100	3
4	BCACAC S504	Statistical Computing and R Programming	Core	3	80	20	100	3
5	BCACAC S505	Cloud Computing	Core	3	80	20	100	3
6	BCACAC S506	Cryptography and Network Security	Core	3	80	20	100	3
7	BCACAP S507	Design and Analysis of Algorithms -Lab	Practical	4	40	10	50	2
8	BCACAP S508	R Programming-Lab	Practical	4	40	10	50	2
9	BCACAP S509	Dynamic Web Programming - Lab	Practical	4	40	10	50	2
10	BCACAS S501	AI Tools and Techniques	Compulsory	2	40	10	50	2
Total				32	640	160	800	26

Semester VI								
Sl. No	Course Code	Title of the Course	Category of Courses	Teaching Hours per Week	SEE	IA	Total Marks	Credits
1	BCACAC S601	Artificial Intelligence and Applications	Core	3	80	20	100	3
2	BCACAC S602	Fundamentals of Data Science	Core	3	80	20	100	3
3	BCACAP S603	Artificial Intelligence-Lab	Practical	4	40	10	50	2
4	BCACAP S604	Fundamentals of Data Science-Lab	Practical	4	40	10	50	2
5	BCACAP S605	Research Methodology+ Project Work	Project Work	22	400	100	500	12
Total				36	640	160	800	22

SEMESTER V

Program Name	BCA-GENERAL	Semester	V
Course Title	Software Engineering (Theory)		
Course Code:	BCACACS501	No. of Credits	03
Contact hours	39 Hours	Duration of SEA/Exam	3 Hours
Formative Assessment Marks	20	Summative Assessment Marks	80

Course Outcomes (COs):

After the successful completion of the course, the student will be able to:

- How to apply the software engineering lifecycle by demonstrating competence in communication, planning, analysis, design, construction, and deployment.
- An ability to work in one or more significant application domains.
- Work as an individual and as part of a multidisciplinary team to develop and deliver quality software.
- Demonstrate an understanding of and apply current theories, models, and techniques that provide a basis for the software lifecycle.
- Demonstrate an ability to use the techniques and tools necessary for engineering practice.

Topics	Hours	Book/Chapter	Page No/Section
UNIT I			
Introduction: The Problem Domain-Software is Expensive, Late and Unreliable, Maintenance and Rework, Software Engineering Challenges-Scale, Quality and Productivity, Consistency and	10	Book 1 Chapter 1	1.1, 1.1.1, 1.1.2, 1.1.3, 1.1.4, 1.2, 1.2.1, 1.2.2, 1.2.3, 1.2.4, 1.3, 1.3.1, 1.3.2 (Page No. 2 - 18)

Function Oriented Design: Design Principles-Problem Partitioning and Hierarchy, Abstraction, Modularity, Top- down and Bottom-up strategies, Module- Level Concepts Coupling, Cohesion, Design Notation and Specification- Structure Charts, Structured Design.		Chapter 6	(Page No. 216 – 232)
UNIT III			
Detailed Design: Detailed Design and PDL- PDL Logic/Algorithm Design, Verification-Design Walkthroughs, Critical Design Review, Consistency Checkers. Coding: Programming Principles and Guidelines-Common Coding Errors, Structured Programming, Some Programming Practices, Coding Standards, Verification-Code Inspections	10	Book 1 Chapter 8 Book 1 Chapter 9	8.1,8.1.1,8.1.2, 8.2,8.2.2,8.2.3 (Page 323-329, 331-333) 9.1,9.1.1,9.1.2,9 .1.3,9.1.4,9.1.5, 9.4,9.4.1 Page No (342-357, 374-376)
UNIT IV			
Testing: Testing Fundamentals- Error, Fault and Failure, Test Oracles, Test Cases and Test Criteria, Black Box Testing- Equivalence Class Partitioning, Boundary Value Analysis, Cause-Effect	09	Book 1 Chapter 10	10.1,10.1.1,10.1 .2, 10.1.3, 10.2, 10.2.1, 10.2.2, 10.2.3, 10.3,10.3.1,10.3 .2, 10.4, 10.4.1,10.4.2, 10.4.3, 10.4.5 (Page 405-410,

Graphing, White Box Testing- Control Flow Based Criteria, Data Flow Based Testing, Testing Process-Levels of Testing, Test Plan, Test Case Specification, Defect Logging and Tracking.			411-418, 424- 431, 439-445, 446-449)
Text Books: 1. Pankaj Jalote, An Integrated Approach to Software Engineering 3rd Edition, Narosa Publishing House. Reference Books: 1. Roger S. Pressman, Software Engineering, A Practitioner's Approach, McGraw Hill 2009. 2. K.K. Aggarwal Yogesh Singh, Software Engineering, 1st Edition, New Age International Pvt Ltd Publishers. 3. Renu Rajni, Software Testin: Methodologies, Tools and Processes, Tata McGraw Hill Education.			

Pedagogy: Lecture/ PPT/ Videos/ Animations/ Quizzes/ Mini
Projects/Group Discussion/ Experiential Learning etc.

Program Name	BCA-GENERAL	Semester	V
Course Title	Dynamic Web Programming (Theory)		
Course Code:	BCACACS502	No.of Credits	03
Contact hours	39 Hours	Duration of SEA/Exam	3 Hours
Formative Assessment Marks	20	Summative Assessment Marks	80

Course Outcomes (COs):

After completing this course satisfactorily, a student will be able to:

- Understand the basics of PHP and MySQL, including their roles in web development.
- Write and execute PHP scripts to create dynamic web pages.
- Use PHP sessions and cookies to manage state and user information.
- Understand the fundamentals of JavaScript, including its syntax and data types.
- Write and execute JavaScript code to create interactive web pages.
- Implement event handling to create responsive user interfaces.

UNIT – I			
Contents	Hours	Books	Page / Sections
Animation: Introduction, Start and End States, Interpolation, Animations in HTML. All About CSS Animations, creating a Simple Animation, Detailed Look at the CSS Animation Property, Keyframes, Declaring Multiple Animations, Wrap-up. All About CSS Transitions, Adding a Transition.	9	1	Page no. (9 – 49)
UNIT – II			
Introduction to PHP: Introduction to PHP, History and Features of PHP, Understanding PHP, Writing Comments in PHP, Sending Data to the Web Browser. Data types in PHP, Keywords in PHP.	10	4 2	Chapter 1(1 – 13)

Using Variables, Constants in PHP, Expressions in PHP, Operators in PHP, Conditional statements: if, if-else, switch, The ? Operator, Looping statements: while Loop, do while Loop, for Loop.		3	Chapter 2(21 - 29) Chapter 3(65 - 72,83,84), Chapter 4 (95 - 123)
UNIT – III			
Arrays in PHP: Creating Arrays, Accessing Array elements, Multidimensional arrays. Indexed v/s Associative arrays, using Array Functions. Functions in PHP: Function definition, Creating and invoking user-defined functions, Formal parameters versus actual parameters, Function and variable scope. Strings in PHP: What is String? Creating and Declaring String, String Functions. Class and Objects in PHP: Creating and accessing a Class &Object, Object properties, object methods.	10	4 2 3 2 2 2	Chapter 2 (54 - 61) Chapter 5 (119) Chapter 6 (173 -179) Chapter 3 (64-74) Chapter 4 (77-86, 92- 96) Chapter 6 (148 -155)
UNIT – IV			
Form Handling: Creating HTML Form, Handling HTML Form data in PHP Database Handling using PHP with MySQL: Introduction to MySQL, Data Types, Accessing MySQL using MySQL Client and Using phpMyAdmin, MySQL Commands, Using PHP with MySQL, PHP MySQL Functions, connecting to MySQL and Selecting the Database, Executing Simple Queries, Retrieving Query Results, Counting Returned Records,	10	3 4 4 4	Chapter 12 (340 -365) Chapter 4 (111 -127) Chapter 5 (130 -151) Chapter 9 (265 -292)

Inserting, deleting and Updating Records with PHP.			
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Textbooks: 1. Animation in HTML, CSS, and JavaScript, Kirupa Chinnathambi, 1st Edition, CreateSpace Independent Pub, 2013. 2. Kevin Tatroe, Peter MacIntyre & Rasmus Lerdorf, Programming PHP, 3rd Edition, O'Reilly 3. Robin Nixon, Learning PHP, MySQL, JavaScript, CSS & HTML5, 3rd Edition, O'Reilly (Soft Copy) 4. Larry Ullman, PHP and MySQL for Dynamic Web Sites, 4th Edition, Peachpit Press Reference Books: 1. Deitel, Paul_Deitel, Harvey_Deitel, Abbey - Internet and World Wide Web How to Program-Pearson Education (US) (2011) 2. Steven Holzner, PHP the Complete Reference 3. Chris Bates - Web Programming_ Building Internet Applications-Wiley (2002) 4. Web Programming with HTML5, CSS, and JavaScript-John Dean.
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Pedagogy: Lecture/ PPT/ Videos/ Animations/ Quizzes/ Mini Projects/Group Discussion/ Experiential Learning etc.

Program Name	BCA-GENERAL	Semester	V
Course Title	Design and Analysis of Algorithms(Theory)		
Course Code:	BCACACS503	No.of Credits	03
Contact hours	39 Hours	Duration of SEA/Exam	3 Hours
Formative Assessment Marks	20	Summative Assessment Marks	80

Course Outcomes (COs):

After the successful completion of the course, the student will be able to:

- Understand the fundamental concepts of algorithms and their complexity, including time and space complexity, worst-case and average-case analysis, and Big-O notation.
- Design algorithms for solving various types of problems, such as Sorting, Searching, and Graph traversal, Decrease-and-Conquer, Divide-and-Conquer and Greedy Techniques.
- Analyze and compare the time and space complexity of algorithms with other algorithmic techniques.
- Evaluate the performance of Sorting, Searching, Graph traversal, Decrease-and-Conquer, Divide-and-Conquer and Greedy Techniques using empirical testing and benchmarking, and identify their limitations and potential improvements.
- Apply various algorithm designs to real-world problems and evaluate their effectiveness and efficiency in solving them.

Contents	Hours	Book	Chapter/Page Nos.
UNIT I			
Introduction: What is an Algorithm? Fundamentals of Algorithmic problem solving, Important Problem Type. Fundamentals of the Analysis of Algorithm Efficiency: Analysis Framework, Measuring the input size, Units for measuring Running time, Orders of Growth, Worst-case, Best-case and Average-case efficiencies. Asymptotic Notations and Basic Efficiency classes - Informal Introduction, O-notation, Ω -notation, θ -notation, Mathematical analysis of non-recursive algorithms - MaxElement, Unique Elements, Mathematical analysis of recursive algorithms- Factorial, Tower of Hanoi	10	1	Chapter 1 1.1 (Page 29-31), 1.2 (Page 35-42), 1.3 (Page 44-49) Chapter 2 2.1 (Page 68-74) 2.2 (Page 78- 81,84-85) 2.3 (Page 87-90) 2.4 (Page 96-101)
UNIT II			
Brute Force & Exhaustive Search: Introduction to Brute Force approach, Selection Sort and Bubble Sort, Sequential search, String Matching, Closest Pair Problem. Exhaustive Search Travelling Salesman Problem, Knapsack Problem.	10	1	Chapter 3, 3.1 (Page 123- 127) 3.2 (Page 130-132) 3.3 (Page 134-135) Study Material
UNIT III			

Exhaustive Search: Depth First Search, Breadth First Search Decrease-and-Conquer: Introduction, Insertion Sort, Binary Search. Divide-and-Conquer: Introduction, Max and Min, Merge Sort, Strassen's Matrix Multiplication.	10	1	Chapter 3 3.5 (Page 148-154) Chapter 4 (Page 157-162) 4.4 (Page 176-178) Chapter 5 (Page 195-200) 5.4 (Page 215-217) Study Material (Max and Min)
Greedy Technique: Introduction, Prim's Algorithm, Kruskal's Algorithm, Huffman Trees. Limitation of Algorithm Power: Lower bound problem, Decision tree. Backtracking: n-queens problem, subset-sum problem	9	1	Chapter 9 (Page 341-348, 351-353, 364-367) Chapter 11 (Page 414-419, 420-423) Study Material
Text Books: 1. Introduction to the Design and Analysis of Algorithms, Anany Levitin: 3rd Edition, Pearson.			
Reference Books: 1. Computer Algorithms/C++, Ellis Horowitz, Satraj Sahni and Rajasekaran, 2nd Edition, 2014, Universities Press. 2. Introduction to Algorithms, Thomas H. Cormen, Charles E. Leiserson, Ronal L. Rivest, Clifford Stein, 3rd Edition, PHI. 3. Design and Analysis of Algorithms, S. Sridhar, Oxford (Higher Education)			

Pedagogy: Lecture/ PPT/ Videos/ Animations/ Quizzes/ Mini Projects/Group Discussion/ Experiential Learning etc.

Program Name	BCA-GENERAL	Semester	V
Course Title	Statistical Computing and R Programming (Theory)		
Course Code:	BCACACS504	No.of Credits	03
Contact hours	39 Hours	Duration of SEA/Exam	3 Hours
Formative Assessment Marks	20	Summative Assessment Marks	80

Course Outcomes (COs):

After the successful completion of the course, the student will be able to:

- Explore fundamentals of statistical analysis in R environment.
- Describe key terminologies, concepts and techniques employed in Statistical Analysis.
- Define Calculate, Implement Probability and Probability Distributions to solve a wide variety of problems.
- Conduct and interpret a variety of Hypothesis Tests to aid Decision Making.
- Understand, Analyze, and Interpret Correlation Probability and Regression to analyze the underlying relationships between different variables.

Contents	Hours	Book / Chapter	Page No
UNIT I			
Introduction of the language, numeric, arithmetic, assignment, and vectors, Matrices and Arrays, Non-numeric Values, Lists and Data Frames.	9	Book 1: Chapter 1 Chapter 2 Chapter 3 Chapter 4 Chapter 5	4-12 17-36 39-58 59-85 89-101

UNIT II			
Reading and writing files, Programming: Calling Functions, Conditions and Loops, stand- alone statement with illustrations in exercise, stacking statements, coding loops, Writing Functions, Exceptions. Basic Data Visualization.	10	Book 1: Chapter 8 Chapter 9 Chapter 10 Chapter 11 Chapter 12 Chapter 14	147-161 165-176 179-212 215-238 241-258 289-306
UNIT III			
Descriptive Statistics: Types of Data, Nominal, Ordinal, Scale and Ratio, Measures of Central Tendency, Mean, Mode and Median, Percentiles, Quartiles, Measures of Variability, Mean Absolute Deviation, Range, Inter-Quartile-Range, Standard Deviation, Z-Scores. Coefficient of Variation, Measure of shapes-Skewness and Kurtosis. Charts: Bar Chart, Pie Chart, Stem and Leaf Diagram. Distributions: Binomial, Poisson, Continuous Probability Distribution, Normal Distribution, Uniform Distribution. Estimating the population mean using the t-distribution.	10	Book 2	1.4(Page No 8-11) 3.1,3.2,3.3, 3.4(Page No 53-84). 2.2(Frequency Polygon - Page No-25, Stem and Leaf -Page No -27). 2.3(Bar Chart, Pie Chart, Page No- 31and 32), Stem and Leaf Plots-Page No-27). 5.1,5.2,5.3, 5.4(Page No 144-164). 6.1,6.2:- (Page No 185-199). 8.2:- (Page No 266-270).
UNIT IV			

Statistical Inference and Hypothesis Testing: Types of Hypotheses, and Sample, Null and Alternate Hypothesis, Level of Significance, Type I and Type II Errors, One Sample t-test, Chi Square Test of Independence, One way ANOVA. Correlation and Regression: Analysis of Relationship, Positive and Negative Correlation, Perfect Correlation, Karl Pearson Coefficient of Correlation, Correlation Matrix, Scatter Plots, Simple Regression Analysis.	10	Book 2	9.1(Page No 296-301, 304-305) 9.3(Page No 314-318) 16.1(Page No 660-668) 11.2(Page No 412-420) 12.1(Page No 472-476)
Textbooks: 1. Tilman M. Davies, “The book of R: A first course in programming and statistics”, San Francisco, 2016. 2. Ken Black, Business Statistics, 7th Edition ,New Delhi, Wiley, 2013. Reference Books: 1. Vishwas R. Pawgi, “Statistical computing using R software”, Nirali Prakashan publisher, e1 edition, 2022. 2. Garrett Grolemond “Hands-On Programming with R” Oreily Publications			

Pedagogy: Lecture/ PPT/ Videos/ Animations/ Quizzes/ Mini Projects/Group Discussion/ Experiential Learning etc.

Program Name	BCA-GENERAL	Semester	V
Course Title	Cloud Computing (Theory)		
Course Code:	BCACACS505	No.of Credits	03
Contact hours	39 Hours	Duration of SEA/Exam	3 Hours
Formative Assessment Marks	20	Summative Assessment Marks	80

Course Outcomes (COs):

After the successful completion of the course, the student will be able to:

- Explain the core concepts of the cloud computing paradigm such as how and why this paradigm shift came about, the characteristics, advantages and challenges brought about by the various models and services in cloud computing.
- Identify resource management fundamentals like resource abstraction, sharing and sandboxing and outline their role in managing infrastructure in cloud computing.
- Analyze various cloud programming models and apply them to solve problems on the cloud.

Contents	Hours	Book / Chapter	Sections
UNIT I			
Introduction Cloud computing at a glance, Historical developments, Building cloud computing environment.	10	Book 1 Chapter 1	1.1 – 1.1.1, 1.1.2, 1.1.3 and 1.1.5; 1.2 – all subsections - 1.2.1 to 1.2.5; 1.3 – including all subsections of 1.3.1 and 1.3.2
UNIT II			
Principles of Parallel and Distributed Computing: Eras of computing, Parallel vs. distributed computing, Elements of parallel computing, Elements of distributed computing, Technologies for distributed computing.	10	Book 1 Chapter 2	2.1 – Complete Section; 2.2 – Complete Section; 2.3 – Subsections 2.3.1, 2.3.2, 2.3.3 and 2.3.4; 2.4 – including all subsections of 2.4.1 to 2.4.4; 2.5 – Subsections 2.5.1 and 2.5.3.
UNIT III			

Virtualization Introduction, Characteristics of virtualized environments, Taxonomy of virtualization techniques, Virtualization and cloud computing Pros and cons of virtualization	10	Book 1 Chapter 3	3.1 – Complete section 3.2- Complete section 3.3- Subsections 3.3.1, 3.3.2 3.4 – Complete section 3.5- Complete section
UNIT IV			
Cloud Computing Architecture Introduction, The cloud reference model, Types of clouds, Economics of the cloud, Open challenges Cloud Applications Scientific applications, Business and consumer Applications	09	Book 1 Chapter 4 Book 1 Chapter 10	4.1- Complete section; 4.2 - Complete section; 4.3 - Complete section; 4.4- Complete section; 4.5 - Complete section; 10.1 -All Sub Sections-10.1.1-10.1.4 10.2 -All Sub Sections-10.2.1-10.2.5
Textbooks: 1. Rajkumar Buyya, Christian Vecchiola, S. Thamarai Selvi: "Mastering Cloud Computing Foundations and Applications Programming", Elsevier, 2013 Reference Books: 1. Barrie Sosinsky: "Cloud Computing Bible", Wiley- India, 2010 2. K Chandrashekar: "Essentials of Cloud Computing", CRC Press, 2015			

3. Derrick Rountree, Ileana Castrillo: “The Basics of Cloud Computing”, Elsevier, 2014
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Pedagogy: Lecture/ PPT/ Videos/ Animations/ Quizzes/ Mini Projects/Group Discussion/ Experiential Learning etc.

Program Name	BCA-GENERAL	Semester	V
Course Title	Cryptography and Network Security (Theory)		
Course Code:	BCACACS506	No.of Credits	03
Contact hours	39 Hours	Duration of SEA/Exam	3 Hours
Formative Assessment Marks	20	Summative Assessment Marks	80

Course Outcomes (COs):

After the successful completion of the course, the student will be able to:

- To understand basics of Cryptography and Network Security
- To be able to secure a message over insecure channel by various means
- To learn about how to maintain the Confidentiality, Integrity and Availability of a data.
- To understand various protocols for network security to protect against the threats in the network

Contents	Hours	Book / Chapter	Section/ Page No
UNIT I			
Computer and Network Security Concepts: Computer Security Concepts, The OSI Security Architecture, Security Attacks, Security Services, Security Mechanisms, Model for Network Security.	10	BOOK 1 Chapter 1	(Pg. No. 21-33, 41-42)
Classical Encryption Techniques: Symmetric Cipher Model. Substitution techniques(caesar cipher), Transposition techniques.		Chapter 3	(Pg. No. 86-91) (92-93) (107-108)
UNIT – II			

Block Ciphers and the Data Encryption Standard: Block Cipher Principles, The Data Encryption (4. 2).	10	BOOK 1 Chapter 4	(Pg. No. 129–131)
Chapter 6		(Pg. No. 172–175,179–180,190-192)	
Chapter 9		(Pg. No. 285–288, 294–296)	
Standard Advanced Encryption Standard: Finite Field Arithmetic (exclude example), AES Structure (exclude Detailed structure), AES Transformation Functions(exclude example), AES Key Expansion (exclude Rationale).			
Public-Key Cryptography and RSA: Principles of Public-Key Cryptosystems, The RSA Algorithm(exclude example).			
UNIT III			
Cryptographic Hash Functions: Applications of Cryptographic Hash Functions, Two Simple Hash Functions, Secure Hash Algorithm (SHA).	10	BOOK 1 Chapter 11	(Pg. No. 341–347, 355–356)
Message Authentication Codes: Message Authentication Requirements, Message Authentication Functions, MACs Based on Hash Functions.		Chapter 12	(Pg. No. 382–384, 391–392,394-395)
Digital Signatures: Digital Signatures, Elgamal Digital Signature Scheme, Schnorr Digital Signature Scheme.		Chapter 13	(Pg. No. 421–426)
Key Management and Distribution: Symmetric Key Distribution Using Symmetric & Asymmetric Encryption.		Chapter 14	(Pg. No. 442–444, 451-452)
UNIT IV			

Transport-Level Security: Web Security Considerations, Transport Layer Security, HTTPS, Secure Shell (SSH), Electronic Mail: Internet Mail Architecture, Email Formats, Email Threats.	09	BOOK 1 Chapter 17	(Pg. No. 547-550, 566-568)
		Chapter 19	(Pg. No. 613-615, 617-620, 625-626)
System security: Intrusion Detection system		BOOK 2 Chapter 21	Pg. 680-681

Text Books:

1. William Stallings, Cryptography and Network Security: Principles and Practices, 7th Edition.
2. William Stallings, Cryptography and Network Security: Principles and Practices, 8th Edition.

Reference Books:

1. Atul Kahate, Cryptography and Network Security, 4th Edition, 2019
2. W. Mao, "Modern Cryptography – Theory and Practice", Pearson Education.

Charles P. Pfleeger, Shari Lawrence Pfleeger – Security in computing – Prentice Hall of India.

Pedagogy: Lecture/ PPT/ Videos/ Animations/ Quizzes/ Mini Projects/Group Discussion/ Experiential Learning etc.

Program Name	BCA-GENERAL	Semester	V
Course Title	Design Analysis and Algorithm -Lab		
Course Code:	BCACAPS507	No.of Credits	02
Contact hours	52 Hours	Duration of SEA/Exam	3 Hours
Formative Assessment Marks	10	Summative Assessment Marks	40

Sl. No.	PART-A Programs
1	Write a program to sort a list of N elements using Selection Sort Technique.
2	Write a program to sort a list of N elements using Insertion Sort Technique.
3	Write a program to read 'n' numbers, find minimum and maximum value in an array using divide and conquer.
4	Write program to implement the BFS algorithm for a graph.
5	Write program to implement the DFS algorithm for a graph.
6	Write a program to implement Strassen's Matrix Multiplication of 2*2 Matrixes.
7	Design and implement in to find a subset of a given set $S = \{S_1, S_2, \dots, S_n\}$ of n positive integers whose SUM is equal to a given positive integer d. For example, if $S = \{1, 2, 5, 6, 8\}$ and $d = 9$, there are two solutions $\{1, 2, 6\}$ and $\{1, 8\}$. Display a suitable message, if the given problem instance doesn't have a solution.

Sl. No.	PART-B
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1	Sort a given set of n integer elements using Merge Sort method and compute its time complexity. Run the program for varied values of n> 5000, and record the time taken to sort.												
2	Write a program to perform Knapsack Problem.												
3	Write a program to perform Travelling Salesman Problem												
4	Write program to generate the Huffman code for the given character and probabilities.<table><tr><td>Character</td><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td></tr><tr><td>Probability</td><td>0.1</td><td>0.1</td><td>0.2</td><td>0.2</td><td>0.4</td></tr></table>	Character	A	B	C	D	E	Probability	0.1	0.1	0.2	0.2	0.4
Character	A	B	C	D	E								
Probability	0.1	0.1	0.2	0.2	0.4								
5	Write a program that implements Prim's algorithm to generate minimum cost spanning Tree.												
6	Write a program that implements Kruskal's algorithm to generate minimum cost spanning tree.												
7	Write program for solving N-queens problem, by taking value of N as input.												

Evaluation Scheme for Lab Examination:

Assessment Criteria		
Program-1	PART-A Writing-5, Logic- 5, Execution -5 Marks	15 Marks
Program-2	PART-B Writing-5, Logic- 10, Execution -5 Marks	20 Marks
Practical Record		05 Marks
Total		40 Marks

Program Name	BCA-GENERAL	Semester	V
Course Title	R Programming -Lab		
Course Code:	BCACAPS508	No.of Credits	02
Contact hours	52 Hours	Duration of SEA/Exam	3 Hours
Formative Assessment Marks	10	Summative Assessment Marks	40

PART A

- Write a program to create a 3 X 3 matrices A and B and perform the following operations.
 - $A^T.B$
 - $B^T.(A.A^T)$
 - $(A.A^T).B^T$
 - $[(B.B^T)+(A.A^T)-100I_3]^{-1}$
- Write R program to find roots of quadratic equation using user defined function. Test the program user supplied values for all possible cases.
- Write R script to generate prime numbers between two numbers using loops
- Write an R program to create a list containing strings, numbers, vectors and logical values and do the following manipulations over the list
 - Access the first element in the list
 - Give the names to the elements in the list
 - Add an element at some position in the list
 - Remove the element
 - Print the first and third element
 - Update the third element
- Write an R program to create a Data Frame with following details and do the following operations.

ItemCode	itemCategory	ItemPrice
1001	Electronics	700
1002	Desktop Supplies	300
1003	Office Supplies	350
1004	USB	400
1005	CD Drive	800

- a) Subset the Data frame and display the details of only those items whose price is greater than or equal to 350.
 - b) Subset the Data frame and display only the items where the category is either “Office Supplies” or “Desktop Supplies”
 - c) Subset the Data frame and display the items where the Itemprice between 300 and 700
 - d) Compute the sum of all ItemPrice
 - e) Create another Data Frame called “item-details” with three different fields itemCode, ItemQtyonHand and ItemReorderLvl and merge the two frames.
6. Create a factor marital_status with levels Married, single, divorced. Perform the following operations on this factor
- a) Check the variable is a factor
 - b) Access the 2nd and 4th element in the factor
 - c) Remove third element from the factor
 - d) Modify the second element of the factor
 - e) Add new level widowed to the factor and add the same level to the factor marital_status
7. Write a R language Script for following operation on Iris Data Set
- a) Load the Iris Dataset
 - b) View first six rows of iris dataset
 - c) Summarize iris dataset
 - d) Display number of rows and columns
 - e) Display column names of dataset.
 - f) Create histogram of values for sepal length
 - g) Create scatterplot of sepal width vs. sepal length
 - h) Create boxplot of sepal width vs. sepal length
 - i) Find Pearson correlation between Sepal.Length and Petal.Length
 - j) Create correlation matrix for dataset

PART-B

Note: Implement Using R Script and Solve Manually

1. Write a R program to create a Vector containing following 8 values and perform the following operations.

4 3 0 5 2 9 4 5

- Find mean, median, mode.
- Find the range.
- Find the 35th and 78th percentile.
- Find the variance and standard deviation
- Find the interquartile range.
- Find the z-score for each value.

[R Script: 8 Marks]

Solution: 12 Marks]

2. Write R script to find the correlation coefficient and type of correlation between advertisement expenses and sales volume using Karl Pearson's coefficient of correlation method (Direct Method).

Firm	1	2	3	4	5	6	7	8	9	10
Advertisement Exp. (Rs. In Lakhs)	11	13	14	16	16	15	15	14	13	13
Sales Volume (Rs. In Lakhs)	50	50	55	60	65	65	65	60	60	50

[R Script: 8 Marks]

Solution: 12 Marks]

3. Write R script to compute the regression equation of y on x from the following data. Predict the value of y when x=7

X	2	4	5	6	8	11
Y	18	12	10	8	7	5

[R Script: 8 Marks]

Solution: 12 Marks]

4. The times taken by a large group of students to complete a piece of homework, T minutes, are Normally distributed with a mean of 57 minutes and standard deviation of 6.5. Find the probability that the time taken by a random student from the group to complete this homework will be less than 60 minutes. Write R

script to Find the probability that the time taken by a random student from the group to complete this homework

a) Will be less than 60 minutes

b) Between 50 and 80 minutes

[R Script: 8 Marks]

Solution: 12 Marks]

5. Write R script to perform the following using binomial distribution

i. If $n=4$ and $p=0.10$, find $P(x=3)$

ii. If $n=12$ and $p=0.45$, find $P(5 \leq x \leq 7)$

[R Script: 8 Marks]

Solution: 12 Marks]

6. A company has three manufacturing plants, and company officials want to determine whether there is difference in the average age of workers at the three locations. The following data are the ages of five randomly selected workers at each plant. Perform a one-way ANOVA to determine whether there is a significant difference in the mean ages of the workers at three plants. Use $\alpha=0.01$. Write R script for the above problem.

Plant(Employee Ages)

1	2	3
29	32	25
27	33	24
30	31	24
27	34	25
28	30	25

[R Script: 8 Marks]

Solution: 12 Marks]

Evaluation Scheme for Lab Examination:

Assessment Criteria		
Program-1	PART-A Writing-5, Logic- 5, Execution-5 Marks	15 Marks
Program-2	PART-B	20 Marks
Practical Record		05 Marks

Total	40 Marks
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Program Name	BCA-GENERAL	Semester	V
Course Title	Dynamic Web Programming -Lab		
Course Code:	BCACAPS509	No.of Credits	02
Contact hours	52 Hours	Duration of SEA/Exam	3 Hours
Formative Assessment Marks	10	Summative Assessment Marks	40

PART A

1. Design a HTML5 web page containing a form with text, password, number, range, email, url, file, submit and reset elements which must be styled using CSS3 according to the following screenshot.

Registration Form

Name:

Password:

Confirm Password:

Contact Number:

Skill Range:

Email:

Profile URL:

Resume:
 No file chosen

2. Design and implement a web page using HTML and CSS to create a box that performs multiple animations simultaneously.
3. Write a PHP program demonstrating the usage of sessions to store and retrieve user credentials like username and password.

Combine the login, welcome, and logout functionalities into two files –‘index.php’ and ‘welcome.php’.

Login

Username:

Password:

Login

Welcome user2

This is a secure area. You're logged in.

[Logout](#)

4. Write a PHP program with a user interface for calculating age based on the user's birth date. Display the age in years, months, and days

Age Calculator

Enter your birth date:

Your age is 19 years, 9 months, and 8 days.

5. Write a PHP program that collects input from the user through a textbox and checks whether the given number is an Armstrong number or not. 1. If the number is indeed an Armstrong number, display all numbers in range from 1 to that specified number (Note : 0 is Not an Armstrong Number). 2. Perform text validation, including checks for positive integers and general text input.

Armstrong Number Checker

Enter a number:

153 is an Armstrong number.

Armstrong numbers from 1 to 153 are: 1 2 3 4 5 6 7 8 9

Armstrong Number Checker

Enter a number:

100 is not an Armstrong number.

Armstrong Number Checker

Enter a number:

Please enter a valid positive integer.

6. Write a PHP program to create an associative array representing a dictionary with words as keys and their meanings as values. Allow users to input a word and display its meaning if found, otherwise,

display a "Word not found" message. (Please keep 10 pairs of keys and their meanings)

Dictionary

Enter a word:

Meaning: A fruit that grows on trees.

Dictionary

Enter a word:

Word not found.

7. Write a PHP program that includes a user form with a text field and submit buttons for various string manipulations. It will display the result accordingly. (for replace, replace 'a' with 'x').

String Manipulation

Enter a string:

Shuffled string: niIda

PART B

1. Design a webpage with a card or image that changes its size, position, and opacity on hover using CSS transitions and transformations. Use different timing functions.
2. PHP program to implement student registration form using Labels, Text Boxes, Text Area, Checkbox, Radio Buttons, Select and Submit button. (First Name, Last Name, Address, E-Mail, Mobile, City, State, Gender, Hobbies, Blood Group). Display user inserted value in a new PHP page in a neat format.
3. Write a PHP program that implements a class to add and find the difference of two distance values given in feet and inches. The user inputs are collected through an HTML form.

Distance Calculator

Distance 1:

Feet:

Inches:

Distance 2:

Feet:

Inches:

Results:

Sum: 9' 5"

Difference: 5' 8"

4. Develop a PHP program that facilitates the addition, multiplication of two matrices. Utilize HTML for the user interface and PHP for the backend logic. Dynamically generate the required number of textboxes based on the specified number of rows and columns. Implement three distinct buttons to perform each matrix operation. For instance, consider the addition of two matrices as an example.

Matrix Addition

Enter the number of rows:

Enter the number of columns:

Matrix A

3	4	6
4	2	-2

Matrix B

5	6	2
3	5	4

Result Matrix

8	10	8
7	7	2

5. Program to implement a login form where users enter their username and password. Validate the credentials against data stored in a MySQL database and grant access if they are correct.
6. Program to create a feedback form with fields for name, email, subject, and message. Store the submitted feedback data in a MySQL database for later review.

Evaluation Scheme for Lab Examination:

Assessment Criteria		
Program-1	PART-A Writing-5, Logic- 5, Execution - 5 Marks	15 Marks
Program-2	PART-B Writing-5, Logic- 10, Execution -5 Marks	20 Marks
Practical Record		05 Marks
Total		40 Marks

Program Name	BCA-General	Semester	V
Course Title	AI Tools and Techniques (Skill Based Practical Course - Compulsory)		
Course Code:	BCACASS501	No. of Credits	02
Contact hours	26 Hours	Duration of SEA/Exam	--
Formative Assessment Marks	10	Summative Assessment Marks	40

Course Outcomes (COs):

At the end of the course, students will be able to:

1. Identify and study different types of AI tools and their environments and develop various projects using them
2. Use the some of the current AI tools.

Q. No	Works to be carried by the students
1	Create an invitation (image format) for the College Annual Day Programme by giving appropriate prompts to AI Tools such as Gemini, ChatGPT, Adobe Firefly or Canva
2	Create a Presentation about any topic using Gamma or NoteBookLM or any other Presentation creation AI Tools
3	Create a static website with at least 5 pages and animated buttons/menu options using Codex, Claude, Caffeine, Gemini, ChatGPT or Antigravity or any other Web Development AI Tool
4	Create a Shopping Cart web Application using Codex, Claude, Caffeine, Gemini, ChatGPT or Antigravity or any other Web Development AI Tool
5	Create a mobile App using AI enabled Android Studio, Replit, Emergent or any other mobile App development AI Tools to detect a car, it's make, model and colour from an image.

Scheme of Evaluation

Internal Assessment	10 Marks (C1) Basis for Internal Assessment Continuous Evaluation: 10 Marks
Evaluation of Report	40 Marks (C2) i) Preparation and presentation: 25 Marks ii) Viva-Voce: 10 Marks iii) Record Book: 05 Marks

SEMESTER VI

Program Name	BCA-GENERAL	Semester	VI
Course Title	Artificial Intelligence and Applications (Theory)		
Course Code:	BCACACS601	No.of Credits	03
Contact hours	39 Hours	Duration of SEA/Exam	3 Hours
Formative Assessment Marks	20	Summative Assessment Marks	80

Course Outcomes (COs):

After the successful completion of the course, the student will be able to:

- Gain a historical perspective of AI and its foundations.
- Become familiar with basic principles and strategies of AI towards problem solving
- Understand and apply approaches of inference, perception, knowledge representation, and learning.
- Understand the various applications of AI

Contents	Hours	Book/Chapter	Sections
UNIT I			
Introduction- What is AI, Foundations of Artificial Intelligence, History of Artificial Intelligence, The State of the Art, Risks and Benefits of AI. Intelligent Agents: Agents and Environments, Good Behaviour: The Concept of Rationality, Nature of Environments, Structure of Agents.	09	Book 1 Chapter 1 Chapter 2	Sections 1.1 to 1.5 Sections 2.1 to 2.4
UNIT II			
Solving Problems by Searching: Problem-Solving Agents, Searching Algorithms, Uninformed Search Strategies, Informed (Heuristic) Search Strategies, Heuristic Functions.	10	Book 1 Chapter 3	Sections 3.1, 3.3, 3.4(3.4.1a and 3.4.3 only),3.5(3.5.1,3.5.2,

			3.5.33.5.6) and 3.6(3.6.1 to 3.6.6)
UNIT III			
Logical Agents: Knowledge –Based Agents, The Wumps World, Logic, Propositional Logic: A Very Simple Logic, Agents Based on Propositional Logic. First Order Logic: Representation Revisited, Syntax and Semantics of First Order Logic, Knowledge Engineering in First-Order Logic. Inference in First-Order Logic: Propositional vs. First-Order Inference, Unification and Lifting, Forward Chaining, Backward Chaining.	12	Book 1 Chapter 7 Chapter 8 Chapter 9	7.1,7.2,7.3 ,7.4,7.7 8.1,8.2.8.3 .8.4 9.1,9.2,9.3 .9.4
UNIT IV			
Machine Learning– Forms of Learning, Supervised Learning, Machine Learning - Decision Trees, Regression and Classification. Applications of AI - Natural Language Processing, Text Classification and Information Retrieval, Speech Recognition , Image processing and computer vision, Robotics	08	Chapter 19	19.1 to 19.6 Study Material
Text Book: 1. Stuart Russel, Peter Norvig: Artificial Intelligence A Modern Approach, 4 th Edition, Pearson Education. Reference Books: 1. Tom Mitchell, “Machine Learning”, 1st Edition, McGraw-Hill,2017 2. Elaine Rich, Kevin Knight, Shivashankar B Nair: Artificial Intelligence, Tata McGraw Hill 3rd edition.			
Pedagogy: Lecture/ PPT/ Videos/ Animations/ Quizzes/ Mini Projects/Group Discussion/ Experiential Learning etc.			

Program Name	BCA-GENERAL	Semester	VI
Course Title	Fundamentals of Data Science (Theory)		
Course Code:	BCACACS602	No.of Credits	03
Contact hours	39 Hours	Duration of SEA/Exam	3 Hours
Formative Assessment Marks	20	Summative Assessment Marks	80

Course Outcomes (COs):

After the successful completion of the course, the student will be able to:

- Understand the concepts of data and pre-processing of data.
- Know simple pattern recognition methods
- Understand the basic concepts of Clustering and Classification
- Know the recent trends in Data Science

Contents	Hours	Book /Chapter	Section
UNIT I			
Data Mining: Introduction, Data Mining as a subject, Definitions, Knowledge Discovery in Databases (KDD) Vs Data Mining, DBMS Vs Data Mining, DM techniques, Problems, Issues and Challenges in DM, DM applications. Data Preprocessing: Why Preprocess the Data? Descriptive Data Summarization, Data Cleaning, Data Integration and Transformation Cleaning, Data and Transformation, Data Reduction.	09	Book 1 Chapter 1 Chapter 3 Book 2 Chapter 2	1.1 and 1.2 3.1 to 3.10 2.1 to 2.5

UNIT II			
Data Warehouse: Introduction-What is Data Warehouse, Difference between Operational Database System and Data Warehouses, Multidimensional Data Model- From Tables and Spreadsheets to Data Cubes, Stars, Snowflakes, and Fact Constellations, OLAP Operations in the Multidimensional Data Model. Data Warehouse Architecture- A Three-Tier Architecture, Data Warehouse Back-End Tools and Utilities, Metadata Repository.	10	Book 2 Chapter 3	3.1, 3.2(3.2.1,3.2.2,3.2.3 and 3.2.6) 3.3(3.3.1 to 3.3.4)
UNIT III			
Mining Frequent Patterns: Basic Concept –Market Basket Analysis, Frequent Item Set Mining Methods, Frequent Pattern Mining: A Road Map. Apriori algorithm, Generating Association Rule, Improving the efficiency of Apriori. Mining Association Rules- Mining Multilevel Association Rules, Mining Multidimensional Association Rules.	10	Book 2 Chapter 5	5.1(5.1.1 to 5.1.3) 5.2(5.2.1,5.2.2,5.2.3) 5.3(5.3.1 and 5.3.2)
UNIT IV			
Classification: Basic Concepts, Issues, Decision Tree Induction, Attribute Selection Measures (Definition and formula -Information gain, Gain ratio, Gini index). Bayesian Classification: Bayes' Theorem, Rule-Based Classification, k Nearest Neighbor. Prediction: Linear Regression, Nonlinear Regression. Cluster Analysis: What Is Cluster Analysis? Types of Data in Cluster	10	Book 2 Chapter 6	6.1 to 6.5 and 6.9.1 6.11

Analysis, Categorization of Major Clustering Methods, Partitioning Methods, Classical Partitioning Methods: k-Means and k-Medoids.			7.1 to 7.4(7.4.1 only)
Text Books: 1. Arun K Pujari – “Data Mining Techniques” 3rd Edition, Universities Press 2. Jiawei Han and Micheline Kambar – “Data Mining Concepts and Techniques” Second Edition Reference Books: 1. Pang-Ning Tan, Michael Steinbach, Vipin Kumar: Introduction to Data Mining, Pearson Education, 2012. 2. K.P.Soman, Shyam Diwakar, V.Ajay: Insight into Data Mining – Theory and Practice, PHI			

Pedagogy: Lecture/ PPT/ Videos/ Animations/ Quizzes/ Mini Projects/Group Discussion/ Experiential Learning etc

Program Name	BCA-GENERAL	Semester	V
Course Title	Artificial Intelligence Lab		
Course Code:	BCACAPS603	No.of Credits	02
Contact hours	52 Hours	Duration of SEA/Exam	3 Hours
Formative Assessment Marks	10	Summative Assessment Marks	40

Sl. No.	PART-A Programs
1	Write a program to simulate a simple vacuum cleaner agent
2	Write a program to implement a rule-based chatbot
3	Write a program to implement Forward Chaining.
4	Write a program to check palindrome using AI search approach.
5	Write a program to sort a list using simple heuristic logic
6	Write a Python program to simulate a goal-based agent that finds a path from a start position to a goal position in a maze.
PART-B	
1	Write a Program to Implement the Greedy Best-First Search algorithm to find a path between two nodes in a graph.
2	Write a Program to implement Tic-Tac-Toe game using Python.
3	Write a Program to Implement 8-Puzzle problem using Python
4	Write a Program to Implement Water-Jug problem using Python.
5	Write a Program to Implement Tower of Hanoi using Python.

6	Write a Python program to represent facts and rules using propositional logic and perform simple inference.
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Evaluation Scheme for Lab Examination:

Assessment Criteria		
Program-1	PART-A Writing-5, Logic- 5, Execution -5 Marks	15 Marks
Program-2	PART-B Writing-5, Logic- 10, Execution -5 Marks	20 Marks
Practical Record		05 Marks
Total		40 Marks

Program Name	BCA-GENERAL	Semester	V
Course Title	Fundamentals of Data Science-Lab		
Course Code:	BCACAPS604	No.of Credits	02
Contact hours	52 Hours	Duration of SEA/Exam	3 Hours
Formative Assessment Marks	10	Summative Assessment Marks	40

Sl. No.	PART-A Programs
1	Write a Python program to perform data preprocessing techniques such as handling missing values, removing noisy data, and transforming datasets.
2	Write a Python program to create and display a simple multidimensional data cube for sales data using rows, columns, and measures.
3	Write a Python program to perform OLAP operations such as roll-up, drill-down, slice, and dice on a sample dataset.
4	Write a Python program to calculate Information Gain, Gain Ratio, and Gini Index for attribute selection in Decision Tree classification.
5	Consider a dataset with varied numerical scales (e.g., Age ranging from 20–60, and Income ranging from ₹20,000–₹150,000). Write Python program to Implement Min-Max Normalization to scale the features strictly between 0 and 1. Also to Implement Z-Score Standardization to centre the data around a mean of 0 with a standard deviation of 1.
6	Write a Python program to perform cluster analysis using the K-Means algorithm and visualize the formed clusters using graphs.
PART-B Programs	
1	Write a Python program to implement the Apriori algorithm for discovering frequent item sets from a transactional dataset and

	display the frequent item sets based on minimum support threshold.
2	Write a Python program to generate Association Rules from frequent itemsets using support, confidence, and lift measures.
3	Write a Python program to implement Decision Tree classification using a suitable dataset and evaluate the classifier performance using accuracy metrics.
4	Write a Python program to implement the Naïve Bayes classifier for predicting class labels from a suitable dataset.
5	Write a Python program to implement K-Nearest Neighbors (KNN) classification using the Iris dataset and classify flower species based on feature values.
6	Write a Python program to implement Linear Regression analysis for predicting student marks based on study hours or similar numerical data.

Evaluation Scheme for Lab Examination:

Assessment Criteria		
Program-1	PART-A Writing-5, Logic- 5, Execution -5 Marks	15 Marks
Program-2	PART-B Writing-5, Logic- 10, Execution -5 Marks	20 Marks
Practical Record		05 Marks
Total		40 Marks

Program Name	BCA-GENERAL	Semester	VI
Course Title	Research Methodology+ Project Work		
Course Code:	BCACAPS605	No.of Credits	12
Contact hours	22 Hours	Duration of SEA/Exam	3 Hours
Formative Assessment Marks	100	Summative Assessment Marks	500

PROJECT GUIDELINES

Preamble: Project work has been made a part of BCA course to give students exposure to Software development exercises. The primary emphasis of the project work is to understand and gain the knowledge of the principles of software engineering practices. As such, during the development of the project students shall involve themselves in all the stages of the software development life cycle (SDLC) like requirements analysis, systems design, software development/coding, testing and documentation, with an overall emphasis on the development of reliable software systems. Since, the project work spans over the entire final semester, the students shall be advised to take up projects for solving problems of software industry or any research organization or the real-life problems suggested by the faculty in- charge of BCA project work in the Institutions. Topic chosen of work must be nontrivial, analytical and application-oriented. It must involve substantial original work and/or development effort based on the theme. Solved, off-the-shelf and pirated work is not entertained. Any attempt of plagiarism or use of unfair means will result in rejection of the work. All activities of the Project Development must be time-bound and the equal participation of the team members expected throughout the Development process.

GENERAL GUIDELINES TO THE INSTITUTIONS

- Calendar of Project Work shall be announced before the commencement of the Sixth semester. Calendar should contain tentative schedules for the submission of Project Proposal, Project Acceptance, Project Synopsis, Problem Analysis Document, System Design Document, Database Design, Detailed Design, Coding and Testing, Final Report, Internal Assessment exams (at least two), Viva/Voce etc.

- Students shall undertake projects with real life problems (that has direct relevance in day-to-day activities or to knowledge extension) either in their Colleges or in industry/research and development laboratories/software companies as recommended by the faculty in-charge of BCA project work in the Institutions. If a student intends to do industry project, the faculty in-charge shall ensure that the projects are genuine and original in nature.
- There shall be not more than three members in a Project team.
- At least two internal assessment exams shall be conducted to evaluate the progress made by the students at different stages of project work. Such exams may include written tests, document verification and presentations, work demonstration, group discussion, viva-voce etc. so as to objectively assess the understanding gained by the students in course of their project work.

PROJECT VALUATION

External and Internal Examiners together conduct project valuation objectively. To begin with, the finer details about various points contained in the scheme of valuation may be conclusively agreed upon through mutual consultation. During project evaluation, a student shall present his/her work through live demonstration of the software application developed as a part of project. However, if live demonstration is not possible due to the reason that some companies do not divulge source code on account of ownership rights or copyrights, students may be allowed to make PPT presentation of their authentic works. In such cases, candidates shall produce necessary declarations issued by the companies to this effect. However, students shall be enabled to present their work in entirety. The primary objective of project evaluation shall be to assess the extent of effort that was put in to meet the objectives of the project and also to gauge the understanding gained by the students in course of their project works. While evaluating Project Reports, examiners shall scrutinize whether Software Development Life Cycle (SDLC) principles have been consistently followed in the project work and the same are documented well in the Reports. However, the relative and overall emphasis of these principles to a particular problem domain chosen may be taken into account so that project evaluations remain fair and objective.

SCHEME OF VALUATION and Marks Distribution

	Particulars	Marks
Internal Assessment:100		
	Progress assessment for TWO times @ 50 marks at each time	100
Project Report Valuation :300 marks		
1	Introduction	20
2	Related studies about the project	20
3	Project plan & implementation-target achieved/output delivered(effectiveness)	250
4	Other mandatory documents & information (certificates, contents, tables, figures, bibliography etc.)	10
Viva-Voce: 100 marks		
1	Live Demonstration (Software execution) or Dry runs (Presentation of authentic screenshots or captured videos may be used to walk through complete scenarios)-consistency and completeness	80
2	Question and Answer (Oral only or Oral and written)	20
Grand Total Marks		500

FORMAT OF PROJECT SYNOPSIS

Synopsis is a brief outline or general view, as of a subject or written work; an abstract or a summary of the Project Work. It must be as brief (NOT MORE THAN 20 A4 sized paper pages) as is sufficient enough to explain the objective and implementation of the project that the candidate is going to take up.

The write up must adhere to the guidelines and should include the following;

1. Title of the Project.
2. Introduction, objectives and scope of the Project.
3. Project category (Database/Web Application/Client-server/Networking/ Multimedia/gaming/Simulation etc.).
4. Tools / Platform, Hardware and Software Requirement specifications.
5. Analysis (DFDs at least up to second level, ER Diagrams/ Class Diagrams, Database Design etc, as per the project requirements).

6. A complete structure which includes: Number of modules and their description to provide an estimation of the student's effort on the project, Data Structures as per the project requirements for all the modules, Process logic of each module, testing process to be used, reports generation (Mention tentative content of report).
7. Whether Industry Defined/Client Defined/User Defined Project Mention the type. Mention the name and Address of the Industry/Client.
8. Limitation of the project.
9. Future scope and further enhancement of the project.

GUIDELINES FOR PREPARATION OF DISSERTATION

1. ORGANISATION OF THE DISSERTATION

The dissertation shall be presented in a number of chapters; starting with Introduction and ending with Conclusion. Each of the chapters will have precise title reflecting the contents of the chapter. A chapter can be subdivided in to sections, sub-sections and sub-sub-section so as to present the content discretely and with due emphasis.

Sequence of items in Dissertation Report

The following sequence may be followed in the preparation of the final dissertation report:

- Cover Page (On the hardboundcover)
- Title Page (Inner Cover Page)
- Certificate from the Institute
- Certificate from the Company
- Declaration
- Acknowledgement
- (Detailed) Table of Contents (with page numbers).
- List of Figures (with figure number, figure titles and page numbers)
- List of Tables with table number, table title and page number.
- Chapters

1. Introduction

a. Introduction of the System

- i. Project Title
 - ii. Category
 - iii. Overview
- b. Background**
 - i.** Introduction of the Company
 - ii.** Brief note on Existing System
- c.** Objectives of the System
- d.** Scope of the System
- e.** Structure of the System
- f.** System Architecture
- g.** End Users
- h.** Software/Hardware used for the development
- i.** Software/Hardware required for the implementation

2. SRS

- a.** Introduction (Brief write-up about SRS)
- b.** Overall Description
 - i.** Product perspective
 - ii.** Product Functions
 - iii.** User characteristics.
 - iv.** General constraints
 - v.** Assumptions
- c.** Special Requirements (Software / Hardware-if any)
- d.** Functional requirement.
 - i.** Module 1
 - ii.** Module 2
- e.** Design Constraints
- f.** System Attributes
- g.** Other Requirements (if any)

3. System Design (Functional Design)

- a.** Introduction (brief write-up about System Design)
- b.** Assumptions and Constraints
- c.** Functional decomposition
- d.** Description of Programs
 - i. Context Flow Diagram (CFD)
 - ii. Data Flow Diagrams (DFDs–Level 0, Level 1, Level 2)
- e.** Description of components
 - i. Functional component 1
 - ii. Functional component 2

4. Database Design (or Data structure)

- a.** Introduction (brief write-up about Database design)
- b.** Purpose and scope

- c. Table Definition
- d. ER diagram

5. Detailed Design (Logic design of modules)

- a. Introduction (brief write-up about Database design)
- b. Structure of the software package (structure chart)
- c. Modular decomposition of the System
 - i. Module1
 - 1. Inputs
 - 2. Procedural details
 - 3. File I/O interfaces
 - 4. Outputs
 - 5. Implementation aspects (if any)
 - ii. Module 2

6. Program code listing

- a. Database connection
- b. Authorization / Authentication
- c. Data store / retrieval / update
- d. Data validation
- e. Search
- f. Named procedures / functions
- g. Interfacing with external devices (if any)
- h. Passing of parameters
- i. Backup/recovery
- j. Internal documentation

7. User Interface (Screens and Reports)

- a. Login
- b. Main Screen / Home page
- c. Menu
- d. Data store / retrieval / update
- e. Validation
- f. View
- g. On screen reports
- h. Data Reports
- i. Alerts
- j. Error messages

8. Testing

- a. Introduction (brief write-up about Software Testing)
 - i. Unit Testing
 - ii. Integrate Testing

iii. System Testing
b. Test Reports

- Conclusion
- Limitations
- Scope for enhancement (future scope)
- Abbreviations and Acronyms (list)
- Bibliography / References (list in specified format)

2. DISSERTATION FORMAT

2.1 Paper

2.1.1 Quality

The dissertation shall be printed on white bond paper, whiteness 95% or above, weight 70 gram or more per square meter.

2.1.2 Size

The size of the paper shall be standard A4; height 297 mm, width 210 mm.

2.1.3 Type-Setting, Text Processing and Printing

The text shall be printed employing Laserjet or Inkjet printer, the text having been processed using a standard text processor. The standard font shall be Times New Roman of 12 pts with 1.5 line spacing.

2.1.4 Page Format

The printed sheets shall have the following writing area and margins:

Top margin .5"

Bottom margin.5"

Left margin1"

Right margin.75"

2.1.5Pageination

Page numbering in the text of the dissertation shall be numerals starting from '**1**' **at the center of the footer**. The text of the written dissertation shall not be less than 60 pages excluding references, tables, questionnaires and other annexure.

Pagination for pages before the Introduction chapter shall be in lower case Roman numerals, e.g., ‘iv’.

2.1.6 Paragraph format

Vertical space between paragraphs shall be about 2.5 line spacing.

The first line of each paragraph should normally be indented by five characters or 12 mm. A candidate may, however, choose not to indent if (s) he has provided sufficient paragraph separation.

A paragraph should normally comprise more than one line. A single line of a paragraph shall not be left at the top or bottom of a page (that is, no windows or orphans should be left). The word at the right end of the first line of a page or paragraph should, as far as possible, not be hyphenated.

2.2 Chapter and Section format

2.2.1 Chapter

Each chapter shall begin number (in Hindu on a fresh page with an additional top margin of about 75 mm. Chapter Arabic) and title shall be printed at the center of the line in 6 mm font size (18 pt) in bold face using both upper and lower case (all capitals or small capitals shall not be used). A vertical gap of about 25 mm shall be left between the chapter number and chapter title lines and between chapter title line and the first paragraph.

2.2.2 Sections and Sub sections

A chapter can be divided into Sections, Sub sections and Sub different concepts separately. Sections and sub-sub sections so as to present sections can be numbered using decimal points, e.g., 2.2 for the second Section in Chapter 2 and 2.3.4 for the fourth Sub Sections and Sub--section in third Section of Chapter 2. Chapters, Sections shall be included in the Contents with page numbers flushed to the right. Further subsections need not be numbered or included in the contents. The Sections and Sub sections titles along with their numbers in 5 and 4mm (16 and 14 pt) fonts, respectively, in bold face shall be flushed to the left (not centered) with 15 mm space above and below these lines. In further subdivisions character size of 3 and 3.5 with bold face, small caps, all caps and italics may be sued for the titles flushed left or centered. These shall not feature in the contents.

2.2.3 Table / Figure Format

As far as possible tables and figures should be presented in portrait style. Small size table and figures (less than half of writing area of a page) should be incorporated within the text, while larger ones may be presented in separate pages. Table and figures shall be numbered chapter wise. For example, the fourth figure in Chapter 5 will bear the number Table Figure 5.4 or Fig.5.4

Table number and title will be placed above the table while the figure number and caption will be located below the figure.

Reference for Table and Figures reproduced from elsewhere shall be cited in the last and separate line in the table and figure caption, e. g. (after McGregor [12]).

3 AUXILIARY FORMATS

1.1 Binding

The dissertation shall be hard cover bound in leather or rexin.

1.2 Front Covers

The front cover shall contain the following details:

- Full title of dissertation in 6 mm 22 point size font properly centered and positioned at the top.
- Full name of the candidate in 4.5 mm 15 point size font properly centered at the middle of the page.
- A 40 mm dia replica of the college emblem followed by the name of the Department and the year of submission, each in a separate line and properly centered and located at the bottom of the page.

1.2.1 Lettering

All lettering shall be embossed in gold.

1.2.2 Bound back

The degree, the name of the candidate and the year of submission shall also be embossed on the bound (side) in gold.

1.3 Blank sheets

In addition to the white sheets (binding requirement) two white shall be put at the beginning and end of the dissertation.

1.4 Title sheet

This shall be the first printed page of the dissertation and shall contain the submission statement: the Dissertation submitted in partial fulfillment of the requirements of the BCA, the name and Roll No. Of the candidate, name (s) of the supervisor and co-supervisor (s) (if any), Department and year of submission.

Questions Paper for Pattern Core Subjects

Duration:3 Hours

Max.Marks:80

Note: Answer any ten Questions from Part-A. And one full Questions from each unit in Part-B

PART A

1. Answer any 10 questions

$10 \times 2 = 20$

- a.
- b.
- c.
- d.
- e.
- f.
- g.
- h.
- i.
- j.
- k.
- l.

PART B

There are 4 Units - UNIT-I, II, III, IV

Each unit contains 2 main questions and it carry 15 Marks. $4 \times 15 = 60$

Each main question contains 2 or more sub questions.

UNIT-I

2.

- a)
- b)
- c)

3.

- a)
- b)
- c)

Same pattern for UNIT – II , UNIT - III and UNIT – IV
